

Lesson 7: Transformations of Functions

Criteria	Novice (1)	Developing (2)	Proficient (3)	Advanced (4)
Identification of Transformations	Incorrectly identifies changes in the graph.	Identifies translations but struggles with reflections or dilations.	Correctly identifies all transformations (shifts, stretches/shrinks, reflections) from an equation.	Correctly identifies all transformations, including complex combinations.
Graphing Transformed Functions	Fails to apply transformations to the parent graph.	Graphs with 1-2 errors in translation or orientation.	Accurately graphs transformed functions, including proper placement of key points.	Consistently graphs complex transformations precisely, using mapping notation.
Writing Equations from Graphs	Unable to determine the equation from a transformed graph.	Corrects for shifts but makes errors in identifying stretch/compression factors.	Correctly writes the equation $g(x) = a \cdot f(b(x-h)) + k$ based on a graph.	Correctly writes equations and justifies the chosen parameters with evidence.
Function Notation & Vocabulary	Cannot distinguish between horizontal and vertical shifts.	Identifies horizontal/vertical changes with some errors.	Corrects uses $f(x) + k$ vs $f(x + k)$ and accurately describes reflections.	Uses precise mathematical language to explain the transformation steps, including sequence.
Technology Usage (e.g., Desmos)	Cannot operate graphing software.	Uses software but cannot match equations to visual transformations.	Accurately uses sliders or input formulas to create intended visual transformations.	Creates complex, creative, or artistic designs using a variety of transformed functions.